MARK SCHEME · TEACHER COPY

12BR · Topic 13: Data Representation

Cambridge A Level Computer Science 9618

Paper 3 Advanced Theory

Total marks: 72

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Question 1 · Oct/Nov 2025 Paper 31, original Q1 (a),(b) [6 marks]

Question	Answer	Marks
1(a)	**One** mark per mark point MP1 `TYPE Vehicle =` MP2 `(M100, M230, T101, T102, T120, T150)` Example answer: `TYPE Vehicle = (M100, M230, T101, T102, T120, T150)`	2

1(b)	**One** mark per mark point
	MP1 `TYPE Booking` and `ENDTYPE` correct
	MP2 Declare used correctly for every field in the response
	MP3 Any four fields correct
	MP4 Remaining fields correct
	
	Example answer:
	`TYPE Booking`
	`  DECLARE BookingNumber : STRING`
	`  DECLARE Destination : STRING`
	`  DECLARE ClientName : STRING`
	`  DECLARE ClientTelephone : STRING`
	`  DECLARE DateOfDeparture : DATE`
	`  DECLARE PickupAddress : STRING`
	`  DECLARE TaxiUsed : Vehicle`
	`ENDTYPE`
		4

Question 2 · Oct/Nov 2025 Paper 31, original Q2 (a),(b) [6 marks]

Question	Answer	Marks
2(a)	**One** mark per mark point MP1 Correct mantissa MP2 Correct exponent **Mantissa** 0 1 1 0 1 0 1 1 1 0 **Exponent** 1 1 1 0 1 0	2

2(b)	**One** mark per mark point for working (**Max 2**) • number converted to binary e.g., positive binary version of 25.3125 = (0)11001.0101 • two's complement version bits flipped and 1 added = 100110.1011 • -32 + 4 + 2 + 0.5 + 0.125 + 0.0625 // -32 + 4 + 2 + 1/2 + 1/8 + 1/16 • movement of binary point seen (5 places) **One** mark per mark point • correct mantissa • correct exponent **Mantissa** 1 0 0 1 1 0 1 0 1 1 **Exponent** 0 0 0 1 0 1	**4**

Question 3 · May/June 2026 Paper 33, original Q1 (a),(b) [5 marks]

Question	Answer	Marks	Guidance
3	Examiners must award marks for each response in line with: • the specific **content** defined in the mark scheme • the specific **skills** defined in the mark scheme or in the level descriptions • the **standard of response** required as exemplified by the standardisation scripts.		
3(a)	**One** mark per mark point (**Max 3**) MP1 The programmer may require a data type for their application that doesn't exist within the programming language MP2 … to represent a particular set/group of data / custom data structure MP3 … under a single identifier MP4 They may wish to extend those data types that are already there using composite data types MP5 … to make the code easier to understand / read MP6 To produce a data type that exactly matches a program's requirements.	**3**	

3(b)	**One** mark per mark point (**Max 2**) MP1 `TYPE Fibonacci =` MP2 `(0, 1, 2, 3, 5, 8, 13, 21)` Example answer `TYPE Fibonacci = (0, 1, 2, 3, 5, 8, 13, 21)`	**2**	Do not allow quotation marks around data elements. The order of the elements must be correct.

3	W	W	0	18	18

Question 4 · May/June 2026 Paper 33, original Q3 (a),(b),(c) [6 marks]

Question	Answer	Marks	Guidance
4	Examiners must award marks **positively**. This means that: • marks are not deducted for errors or omissions • marks are awarded for correct/valid answers as defined in the mark scheme and also for **other valid answers** which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate • the quality of spelling, punctuation and grammar are judged **only** when the mark scheme indicates that these features are **specifically assessed** in the question. However, the meaning of all responses must be unambiguous.		
4(a)	**One** mark for an appropriate use of serial file organisation (**Max 1**) Batch processing tasks Creating a transaction file for a batch Storing bills/payroll Backup Appropriate example where data is added chronologically	**1**	
4(b)	**One** mark per mark point (**Max 2**) MP1 Records are stored in a file one after the other MP2 … in chronological order MP3 New records are appended/added at the end of the file	**2**	
4(c)	**One** mark per mark point (**Max 3**) MP1 Sequential file access is used MP2 Each record in the file is searched / checked, one after the other, MP3 … until the end of the file is reached MP4 … until the record is found.	**3**	

4	W	N2	4	14	18

 Total: 72 marks

Question 5 · May/June 2026 Paper 33, original Q5 (a),(b) [7 marks]

Question	Answer	Marks	Guidance
5	Examiners must use the **full range of marks** defined in the mark scheme, unless limited by the quality of the candidate responses seen.		
5(a)	**One** mark per mark point for working (**Max 2**) • number converted to binary e.g., positive binary version of 18.5625 = 010010.100100 • two's complement version bits flipped and 1 added = 101101.011100 **One** mark per mark point (**Max 2**) • correct mantissa • correct exponent **Mantissa**: 1 0 1 1 0 1 0 1 1 1 0 0 **Exponent**: 0 1 0 1	**4**	Allow alternative methods of working.
5(b)	**One** mark per mark point (**Max 3**) MP1 If the two numbers were added together, the value would be 214.75 MP2 This value exceeds the highest/largest number that can be stored in this system with an exponent of 7, (which is 127.9375) MP3 An exponent of 8 would be required to increase the range of possible numbers beyond 128 MP4 Overflow would occur	**3**	

5	W⇨N2	N5	4+7=11	7	18

Question 6 · May/June 2026 Paper 32, original Q4 (a),(b),(c) [6 marks]

Question	Answer	Marks	Guidance
6	Examiners must **consistently** apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.		
6(a)	**One** mark for an appropriate use of sequential file organisation (**Max 1**) Batch processing tasks Recording log files Generating bills/payroll Backup A valid example where the data is **stored in a specified order**	**1**	Do not allow examples that simply describe sequential file organisation, such as 'records are stored one after the other'.

6(b)	**One** mark per mark point (**Max 2**) MP1 Records are stored in a file one after the other / linearly MP2 … in a given order MP3 … usually based on a key field / unique identifier	**2**	Chronological order is not correct for MP2 – that would be serial organisation.
6(c)	**One** mark per mark point (**Max 3**) MP1 Each record in the file is searched, one after the other / linearly // Each record in the file is searched from the physical start of the file MP2 … using the record key MP3 … until the record is found. MP4 … until the value of the record key being searched has been exceeded // … until the end of the file is reached.	**3**	Candidates whose answer is about searching an index as for index sequential files do not answer this question.

Question 7 · Oct/Nov 2025 Paper 32, original Q1 (a),(b) [6 marks]

Question	Answer	Marks

7(a)	**One** mark for each mark point
	MP1 Use of correct `Flight1` variable with given field names
	MP2 Correct assignments of four string data values
	MP3 Correct assignment of date data value
	
	Example answer
	`Flight1.FlightNumber ← "SB2789"`
	`Flight1.Destination ← "Dublin"`
	`Flight1.FlightDate ← 30/07/2025`
	`Flight1.Gate ← "N03"`
	`Flight1.Airline ← "Cambridge Airways"`
		3

7(b)(i)	**One** mark per mark point MP1 `TYPE GateID =` MP2 `(N01, N02, N03, W01, W02, W03, W04)` Example answer `TYPE GateID = (N01, N02, N03, W01, W02, W03, W04)`	**2**
7(b)(ii)	`DECLARE Gate : GateID`	**1**

Question 8 · Oct/Nov 2025 Paper 32, original Q2 (a),(b) [6 marks]

Question	Answer	Marks
8(a)	**Two** marks for working • number converted to binary (0)1111100.0111 • use of exponent = 7 // Moving binary point the correct number (7) of places **One** mark for correct answer **Mantissa**: 0 1 1 1 1 1 0 0 0 1 1 1 **Exponent**: 0 1 1 1	**3**

8(b)	**Two** marks for working • correct use of exponent seen • correct conversion method from binary to denary **One** mark for correct answer Working: 1010001.01011 // moving bp 6 places to right Evaluation of two's complement −64 +16 + 1 + 0.25 + 0.0625 + 0.03125 // −64 + 16 + 1 + 1/4 + 1/16 + 1/32 // Converting two's complement back and evaluating positive binary number 32 + 8 + 4 + 2 + 0.5 + 0.125 + 0.03125 // 32 + 8 + 4 + 2 + 1/2 + 1/8 + 1/32 Fractions methods - award both working marks for either (– 2048 + 512 + 32 + 8 + 2 + 1) / 2048 x 2^6 = -1493 / 32 **OR** – 1 + 1/4 + 1/64 + 1/ 256 + 1/1024 + 1/2048 x 2^6 = -1493 / 32 Answer: −46.65625 // $-46^{21}/_{32}$	**3**

Question 9 · May/June 2026 Paper 31, original Q2 (a),(b),(c) [6 marks]

Question	Answer	Marks	Guidance
9	Examiners must award **whole marks** (not half marks, or other fractions).		
9(a)	**One** mark for an appropriate use of random file organisation (**Max 1**) Booking systems e.g. airline Operating system file management systems Video games Encryption utilities Compression utilities	**1**	Do not allow DBMS. Do not allow definitions of random file organisation.

9(b)	**One** mark per mark point (**Max 2**) MP1 Records are stored in no particular order / any available position MP2 A record key is used to determine the location where the record is to be stored MP3 … which is generated using a **hash** function on the key field MP4 New records are added in an empty location	**2**	Do not allow use of data/file instead of record Do not accept random as the description.
9(c)	**One** mark per mark point (**Max 3**) MP1 Direct file access is used MP2 A hashing function is used on the key field to find the location in the file where the record is stored MP3 If the required record is found at the hashed location it is retrieved MP4 If the required record is not found, a collision resolution strategy is employed // Records from that point are searched linearly until the required record is found // Linear probing	**3**	Do not allow use of data/file instead of record Allow open or closed hashing / addressing for MP4.

Question 10 · May/June 2026 Paper 31, original Q3 (a),(b) [6 marks]

Question	Answer	Marks	Guidance
10	Examiners must award marks **positively**. This means that: • marks are not deducted for errors or omissions • marks are awarded for correct/valid answers as defined in the mark scheme and also for **other valid answers** which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate • the quality of spelling, punctuation and grammar are judged **only** when the mark scheme indicates that these features are **specifically assessed** in the question. However, the meaning of all responses must be unambiguous.		

10(a)	**One** mark per mark point (**Max 3**) MP1 correct **method** to find the binary number MP2 correct **use** of exponent MP3 correct answer in the space provided Working: –10.8125 converted to binary positive binary version of 10.8125 = 01010.1101 // two's complement version - bits flipped and 1 added = 10101.0011 // -8+2+0.5+0.25+0.0625 // -16+4+1+0.125+0.0625. Movement of binary point by 4 places **Mantissa**: 1 0 1 0 1 0 0 1 1 0 **Exponent**: 0 0 0 1 0 0	**3**	Allow other methods of working
10(b)	**Two** marks for working (**Max 2**) • correct application of exponent seen • correct method to find the final answer Working: 10100001.11 (moving bp 7 places to right) Evaluation of two's complement −128 +32 + 1 + 0.5 + 0.25 **One** mark for correct answer (**Max 1**) Denary value: −94.25 // $-94^{1}/_{4}$	**3**	Allow other methods of working

Question 11 · Oct/Nov 2025 Paper 33, original Q1 (a),(b) [6 marks]

Question	Answer	Marks

Total: 72 marks

11(a)	**One** mark per mark point
	MP1 `TYPE VideoLibrary` and `ENDTYPE` correct
	MP2 Declare used correctly for every field in the response
	MP3 All Three STRING fields correct (shaded)
	MP4 Remaining Three fields correct (unshaded)
	
	Example answer
	`TYPE VideoLibrary`
	`    DECLARE VideoID : STRING`
	`    DECLARE Title : STRING`
	`    DECLARE ReleaseYear : INTEGER`
	`    DECLARE PurchaseDate : DATE`
	`    DECLARE VideoFormat : STRING`
	`    DECLARE RunningTime : INTEGER`
	`ENDTYPE`
		4

11(b)	**One** mark for identification of field and **one** mark for reason `VideoFormat` This field can have a fixed range of possible values	**2**

11	15	2	3	7	11	15	17	19	23	0	0

Question 12 · Oct/Nov 2025 Paper 33, original Q2 (a),(b) [6 marks]

Question	Answer	Marks
12(a)	**One** mark for two's complement version and **one** mark for denary version **Mantissa** 0 1 1 1 1 1 1 1 **Exponent** 0 1 1 1 1 1 1 1 127×2^{120} // 0.9921875×2^{127} // $127/128 \times 2^{127}$	**2**

Total: 72 marks

12(b)	**One** mark per mark point for working (**Max 2**) • number converted to binary e.g., positive binary version of 3.59375 = (0)11.10011 • negative two's complement version - bits flipped and 1 added = 100.01101 • -4 + 1/4 + 1/8 + 1/32 // -4 + 025 +0.125 + 0.03125 // -(64 + 32 + 16 + 2 + 1)/32 • $-2^2 + 2^{-2} + 2^{-3} + 2^{-5}$ • $2^2 (- 2^0 + 2^{-4} + 2^{-5} + 2^{-7})$ **One** mark per mark point • correct mantissa • correct exponent, with working seen. **Mantissa** 1 0 0 0 1 1 0 1 **Exponent** 0 0 0 0 0 0 1 0	**4**

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